

Programme: B.Sc.DS

Division: _____

Roll No. _____

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date: 02/01/2026

Mid Term Exam (January 2026)

Time: 1hr

TYB.Sc.DS SEM-VI

Subject: Machine Learning

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	OC	Q.1	Attempt any 2 questions.	Marks 10
2	1	a.	What types of problems are suitable for Machine Learning techniques?	5
4	1	b.	Draw and explain the general architecture of a Machine Learning system.	5
2	1	c.	Explain the concepts of generalization, bias, and variance in ML models.	5
1	1	d.	Define reinforcement learning and explain its working with a real-world example.	5
Bloom's Taxonomy Level	OC	Q.2	Attempt any 2 questions..	Marks 10
5	2	a.	Differentiate between binary, multi-class, and multi-label classification problems.	5
2	2	b.	Explain the need and working principle of the K-Nearest Neighbor (KNN) algorithm.	5
3	2	c.	How is entropy used in attribute selection in Decision Trees?	5
4	2	d.	Compare Information Gain and Gini Index in terms of their usage.	5

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VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date: 05/01/2026

Mid Term Exam (Jan 2026)

Time: 1hr

TYB.Sc.DS SEM-VI

Subject: Internet of Things: Theory

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	OC	Q.1	Attempt any 2 questions.	Marks 10
1,2	1	a.	Define and Explain Ubiquitous Computing.	5
2,3	1	b.	Explain calm and ambient technology using example of Live wire.	5
4,5	3	c.	"Data available through IOT device belongs to public or company which implements the IOT device". Discuss.	5
1,2	2	d.	Define protocol. Explain the following application layer protocols: HTTPS, SMTP, FTP, POP3, IMAP.	5
Bloom's Taxonomy Level	OC	Q.2	Attempt any 2 questions.	Marks 10
4	3	a.	What are the challenges when we move from prototype to mass production?	5
4,5	3	b.	Discuss open source versus closed source hardware and software. State their advantages and disadvantages.	5
2	1	c.	Write short note on Raspberry Pi.	5
2,3	1	d.	Explain the following IOT devices built with Arduino. (i) The Good Night Lamp (ii) Botanicals (iii) Baker Treat	5

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VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date: 06/01/2026

Mid Term Exam (Jan 2026)

Time: 1hr

TYB.Sc.DS SEM-VI

Subject: Applied Business Analytics

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	OC	Q.1	Attempt any 2 questions.	Marks 10
3	1	a.	What is spreadsheet modelling? Explain.	5
3	1	b.	What are the measures of shape? Explain.	5
4	1	c.	Differentiate between one-way table & two-way table in what-if analysis.	5
4	1	d.	Explain in brief the difference between outlier and missing value.	5
Bloom's Taxonomy Level	OC	Q.2	Attempt any 2 questions.	Marks 10
2	2	a.	How to import data into Excel with Power Query?	5
2	2	b.	What is DAX? Explain with an example.	5
3	2	c.	How to deal with Data with Errors?	5
4	2	d.	Differentiate between Subjective and Objective Probabilities.	5

OC No.	Particulars
OC1	To understand basics of statistical concepts like probability distribution, hypothesis testing etc.
OC2	To experiment with Business Intelligence Tools for Data Analysis.

Programme: B.Sc.IT

Division: _____

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VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date: 07/01/2026

Mid Term Exam (Jan 2026)

Time: 1 hr

TYB.Sc.DS SEM-VI

Subject: Healthcare Analytics

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	OC	Q.1	Attempt any 2 questions.	Marks																
2	1	a.	Write a note of Computer science domain in Healthcare Analytics.	5																
2	1	b.	Which software is required for Healthcare Analytics?	5																
4	1	c.	Write a note on "Patient data – the journey from patient to computer".	5																
4	1	d.	Illustrate the use of "Review of Systems" in Healthcare Analytics.	5																
	OC	Q.2	Attempt any 2 questions.																	
2	2	a.	Write a note on International Classification of Disease (ICD).	5																
4	2	b.	Break down healthcare analytics into different categories.	5																
2	2	c.	Write about the different data formats used in Healthcare.	5																
6	2	d.	The 2X2 contingency matrix is given for chest pain and myocardial infarction for 500 patients. Calculate the sensitivity, specificity and likelihood ratios for chest pain (+ and -). <table><tr><td></td><td>Myocardial infarction present (D+)</td><td>Myocardial infarction present (D-)</td><td>Total</td></tr><tr><td>Chest pain present (T+)</td><td>45 (TP)</td><td>120 (FP)</td><td>165</td></tr><tr><td>Chest pain present (T-)</td><td>25 (FN)</td><td>310 (TN)</td><td>335</td></tr><tr><td></td><td>70</td><td>430</td><td>500</td></tr></table>		Myocardial infarction present (D+)	Myocardial infarction present (D-)	Total	Chest pain present (T+)	45 (TP)	120 (FP)	165	Chest pain present (T-)	25 (FN)	310 (TN)	335		70	430	500	5
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VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

Date: 03/01/2026

Mid Term Exam (Jan 2026)

Time: 1 hr

TYB.Sc.DS SEM-VI

Subject: Exploratory Data Analysis

Marks: 20

Instructions: All questions are compulsory.

Bloom's Taxonomy Level	OC	Q.1	Attempt any 2 questions.	Marks 10
2	1	a.	Compare Cross Sectional, times series and panel data giving appropriate examples	5
2	1	b.	State the differences between statistic and parameter giving examples	5
4	1	c.	Highlight the importance of data in EDA.	5
6	1	d.	i. A university wants to track: a) The proportion of students who access the LMS daily b) The maximum marks scored in a final exam c) The true average CGPA of all students Identify whether each is a point estimate, interval estimate, or parameter, and justify ii. A college wants to predict success probability of students in placements using attendance %, CGPA, English level rating, LinkedIn activity logs. a) Identify the correct measurement scale for each variable. b) Can mean be used for all variables? If not, explain. c) Decide which variable needs normalization and why.	5
Bloom's Taxonomy Level	OC	Q.2	Attempt any 2 questions.	Marks 10
2	2	a.	Explain with the help of an example, how run sequence plot helps in drawing insights from data.	5
4	2	b.	State the limitations of quantitative techniques. Which type of representation can overcome the limitations? Explain.	5
2	2	c.	State the four underlying assumptions of EDA.	5
6	2	d.	Compare EDA with Classical data analysis in brief. A hospital wants to understand the average patient waiting time and decide whether delays are acceptable. The hospital wants to address the following questions: i) Is the mean waiting time significantly higher than 30 minutes? ii) How are waiting times distributed and where are the bottlenecks? Identify which of Classical and Exploratory Data Analysis can be used to answer the questions. Justify your answer.	5

OC No.	Particulars
OC1	Understand importance of data and its types in Exploratory Data Analysis
OC2	Classify EDA and summary statistics in context of interpretation